

## ORIGINAL RESEARCH

# Impact of Left Atrial Dilatation on Catheter Ablation Outcomes for Persistent Atrial Fibrillation

## Does Size Matter?

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## ABSTRACT

**BACKGROUND** Left atrial (LA) enlargement is commonly used in patient selection for catheter ablation. However, its role in predicting postablation outcomes in persistent atrial fibrillation (AF) remains uncertain.

**OBJECTIVES** The study goal was to evaluate the relationship between baseline LA volume index (LAVI) and arrhythmia outcomes after ablation for persistent AF.

**METHODS** A total of 338 patients with persistent AF enrolled in the CAPLA (Catheter Ablation for Persistent Atrial Fibrillation: Pulmonary Vein Isolation vs. PVI plus Posterior Wall Isolation) trial were screened for pre-ablation LAVI evaluation on echocardiography. Atrial arrhythmia recurrence and burden were assessed at 1 year and 3 years' postablation using intensive rhythm monitoring (implantable devices, twice-daily mobile electrocardiograms, or serial Holter monitors). The association between LAVI and arrhythmia recurrence, burden, and recurrence phenotype were analyzed by using both survival and logistic regression models.

**RESULTS** A total of 281 (83.1%) of 338 patients had baseline LAVI data available. At 1 year, arrhythmia recurrence occurred in 130 (46.3%) patients, increasing to 173 (61.6%) at 3 years. LAVI was not associated with arrhythmia recurrence in a multivariable analysis (OR: 1.01; 95% CI: 0.99-1.02;  $P = 0.55$ ) nor when stratified according to LAVI category (log-rank test,  $P = 0.19$ ). However, among patients with arrhythmia recurrence, increased LAVI was associated with greater arrhythmia burden ( $R^2 = 0.12$ ;  $P < 0.001$ ) and persistent (vs paroxysmal) recurrence (OR: 1.78 per 10 mL/m<sup>2</sup>; 95% CI: 1.33-2.47;  $P < 0.001$ ) at 12 months.

**CONCLUSIONS** In persistent AF, baseline LAVI did not predict arrhythmia recurrence after first-time ablation but was associated with higher AF burden and persistent recurrence phenotype at 1 year postablation. These findings suggest that LA size may influence recurrence burden and phenotype rather than overall ablation success. (JACC Clin Electrophysiol. 2026;■:■-■) © 2026 by the American College of Cardiology Foundation.

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ABBREVIATIONS  
AND ACRONYMS

AF = atrial fibrillation

AFEQT = Atrial Fibrillation  
Effect on Quality-of-Life  
questionnaire

ECG = electrocardiogram

LA = left atrial

LAVI = left atrial volume index

LV = left ventricular

LVEF = left ventricular  
ejection fraction

PWI = posterior wall isolation

TTM = transtelephonic  
monitoring

Catheter ablation has emerged as a cornerstone therapy for rhythm control in patients with persistent atrial fibrillation (AF), offering superior outcomes to antiarrhythmic drugs in terms of symptom relief, health care utilization, and quality of life.<sup>1,2</sup> Despite advances in ablation techniques and technology, arrhythmia recurrence after ablation remains common, particularly in patients with persistent AF.<sup>3</sup> Identifying predictors of recurrence remains essential to optimize patient selection and procedural strategy.

Left atrial (LA) enlargement has long been associated with atrial arrhythmogenesis and has been proposed as a key determinant of patient selection for catheter ablation.<sup>4</sup> Meta-analyses and small observational cohorts have linked LA volume index (LAVI) with recurrence,<sup>5,6</sup> but these findings are limited by heterogeneity in imaging modalities, inconsistent thresholds for LA dilation, and variable rhythm-monitoring strategies. Furthermore, available evidence is weighted heavily toward paroxysmal AF, with data in persistent AF comparatively sparse. In addition, studies to date have not examined the relationship between LA size and arrhythmia burden, with existing analyses limited to binary arrhythmia recurrence.

The CAPLA (Catheter Ablation for Persistent Atrial Fibrillation: Pulmonary Vein Isolation vs. PVI plus Posterior Wall Isolation) study enrolled patients with persistent AF undergoing first-time radiofrequency ablation.<sup>7</sup> The trial featured structured, long-term follow-up for up to 3 years' postablation, including intensive rhythm monitoring, to allow quantification of the atrial arrhythmia burden. The aim of the current substudy was to assess the relationship between baseline LA size and ablation success in the CAPLA cohort, using both binary recurrence and AF burden.

## METHODS

This was a post hoc substudy of the CAPLA trial (ACTRN12616001436460), a multicenter, international, investigator-initiated randomized controlled study evaluating ablation strategies for persistent AF. The full trial protocol has been published previously.<sup>7,8</sup> In brief, CAPLA enrolled 338 patients with symptomatic persistent AF undergoing first-time radiofrequency catheter ablation. Persistent AF was defined as a continuous AF episode  $\geq 7$  days and  $< 3$  years in duration. Patients with paroxysmal and very long-standing persistent AF ( $\geq 3$  years), hypertrophic cardiomyopathy, or prior AF ablation were excluded.

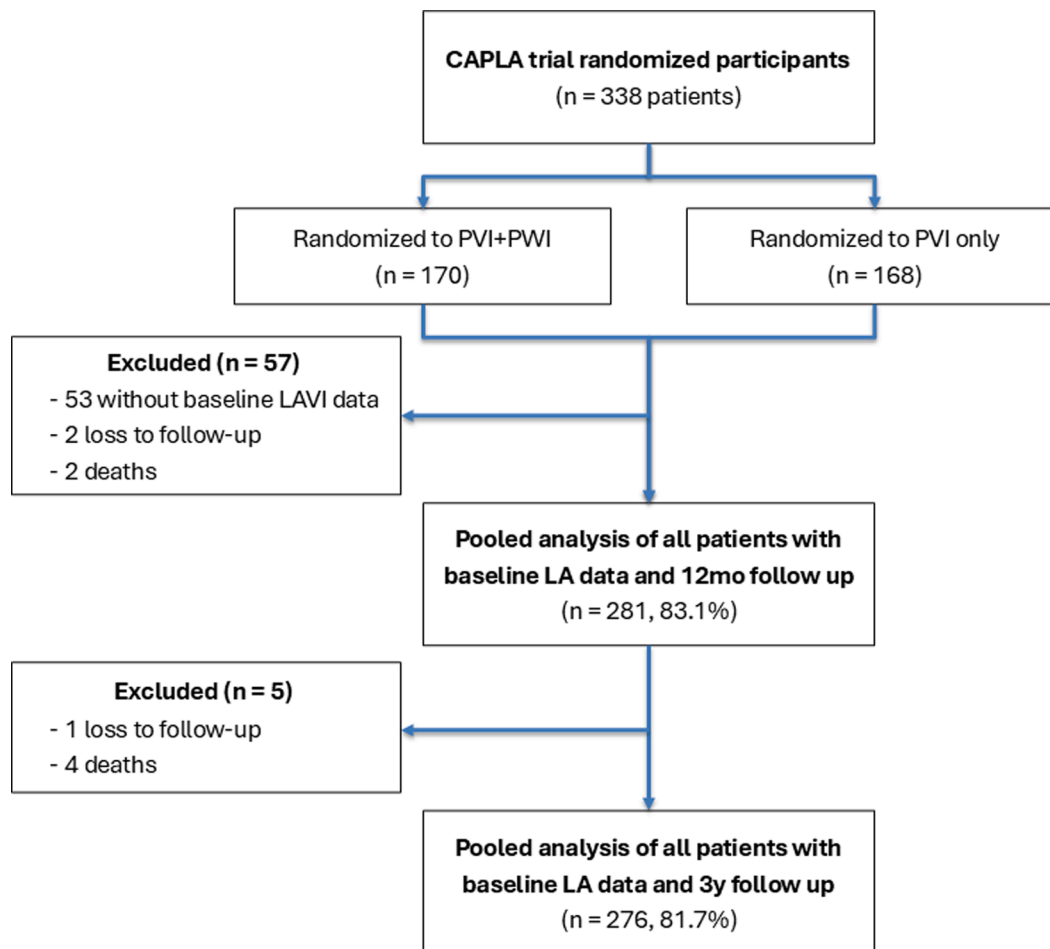
An upper limit of LA size was not an exclusion criterion.

All patients provided written informed consent, and the study was approved by the Alfred Hospital Ethics Committee and the ethics boards of all participating institutions. For the current analysis, we included all patients with available baseline echocardiographic data to calculate LAVI.

**ECHOCARDIOGRAPHIC ASSESSMENT.** Baseline transthoracic echocardiography was performed in all participants in the most recent 3 months preceding their index ablation. LA volume was measured by using the biplane disk summation method from the apical 2- and 4- chamber windows and indexed to body surface area to yield LAVI. Based on guideline-recommended thresholds,<sup>9</sup> LA dilation was categorized as: normal ( $\leq 34$  mL/m<sup>2</sup>), mild (34.1–42 mL/m<sup>2</sup>), moderate (42.1–48 mL/m<sup>2</sup>), or severe ( $> 48$  mL/m<sup>2</sup>). Left ventricular (LV) ejection fraction (LVEF) was calculated by using the Simpson biplane method using the 2- and 4- chamber windows.

**ABLATION PROCEDURE AND FOLLOW-UP.** All patients underwent catheter ablation under general anesthesia using three-dimensional electro-anatomical mapping and irrigated contact force-sensing catheters. Pulmonary vein isolation was performed using wide antral circumferential ablation. Patients randomized to posterior wall isolation (PWI) received linear ablation across the roof and floor of the left atrium, with ablation inside the box when necessary to achieve entrance and exit block. Antiarrhythmic drugs were discontinued at the end of the 90-day blanking period unless recurrence occurred.

Patients underwent continuous or scheduled arrhythmia monitoring for up to 3 years' postablation using 1 of 3 modalities: 1) implantable cardiac electronic devices; 2) implantable loop recorders with monthly remote transmissions; or 3) transtelephonic monitoring (TTM) via twice-daily handheld electrocardiograms (ECGs) with the KardiaMobile device (AliveCor). Patients nonadherent to handheld monitoring underwent 24-hour Holter monitoring at 3, 6, 9, and 12 months. Recurrence was defined as any documented AF, atrial flutter, or atrial tachycardia lasting  $\geq 30$  seconds after a 90-day blanking period. AF burden was calculated as the percentage of AF over the total monitored time, derived from device diagnostics where available, from cumulative onset-offset times on handheld ECGs, or as the proportion of arrhythmia during Holter monitoring. Burden at 12 months was assessed from all available data in the first year postablation, whereas burden at 3 years was

**FIGURE 1** CONSORT Diagram

Consolidated Standards of Reporting Trials (CONSORT) diagram for the current substudy of the CAPLA trial. LA = left atrial; LAVI = left atrial volume index; PVI = pulmonary vein isolation; PWI = posterior wall isolation.

assessed by using data collected between 2.5- and 3.5-years' postablation.<sup>10</sup> Recurrence phenotype (paroxysmal vs persistent) was adjudicated during the first year postablation; standardized phenotyping was not available at 3 years, and therefore longer term phenotype analyses were not performed. All rhythm monitoring data were adjudicated by a blinded central monitoring committee.

Quality-of-life was assessed by using the Atrial Fibrillation Effect on Quality-of-Life questionnaire (AFEQT), administered at the preablation, 12-month, and 3-year time points.

**STATISTICAL ANALYSIS.** Continuous variables are presented as mean  $\pm$  SD or median (Q1-Q3). Categorical variables are presented as frequencies and

percentages. Time-to-recurrence analysis was performed by using Kaplan-Meier survival estimates and Cox proportional hazards models over 3-year follow-up; death and loss to follow-up were treated as censoring events. Logistic regression was used to evaluate the association between LAVI (as a continuous variable) and arrhythmia recurrence at 12 months in a multivariable model adjusted for clinically relevant covariates: age, sex, body mass index, LVEF, AF duration, hypertension, and obstructive sleep apnea. Bivariate continuous data were analyzed first using linear regression; linearity was confirmed by comparison with higher order models.

All analyses were conducted by using R version 4.2.2 (R Foundation for Statistical Computing).

**TABLE 1 Cohort Characteristics (N = 281)**

|                                                                                                                                                                                                                                                                     | Value         |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| <b>Demographic characteristics and comorbidities</b>                                                                                                                                                                                                                |               |
| Age, y                                                                                                                                                                                                                                                              | 64.8 ± 9.8    |
| Male                                                                                                                                                                                                                                                                | 218 (77.6)    |
| BMI, kg/m <sup>2</sup>                                                                                                                                                                                                                                              | 29.9 ± 5.3    |
| Obesity (BMI >27 kg/m <sup>2</sup> )                                                                                                                                                                                                                                | 189 (67.3)    |
| Hypertension                                                                                                                                                                                                                                                        | 134 (47.9)    |
| Congestive heart failure                                                                                                                                                                                                                                            | 124 (44.1)    |
| Ischemic heart disease                                                                                                                                                                                                                                              | 33 (11.7)     |
| Type 2 diabetes mellitus                                                                                                                                                                                                                                            | 26 (9.3)      |
| Previous stroke                                                                                                                                                                                                                                                     | 16 (5.7)      |
| Obstructive sleep apnea                                                                                                                                                                                                                                             | 54 (19.2)     |
| Alcohol excess <sup>a</sup>                                                                                                                                                                                                                                         | 33 (11.7)     |
| <b>Echocardiographic characteristics</b>                                                                                                                                                                                                                            |               |
| Left atrial volume index, mL/m <sup>2</sup>                                                                                                                                                                                                                         | 47.9 ± 16.4   |
| Left ventricular ejection fraction, %                                                                                                                                                                                                                               | 52.6 ± 12.3   |
| LVEDD, mm                                                                                                                                                                                                                                                           | 51.7 ± 7.4    |
| E/e'                                                                                                                                                                                                                                                                | 10.3 ± 4.0    |
| Right ventricular systolic pressure, mm Hg                                                                                                                                                                                                                          | 26.9 ± 7.3    |
| <b>AF characteristics</b>                                                                                                                                                                                                                                           |               |
| Time since AF diagnosis, mo                                                                                                                                                                                                                                         | 13 (6.3-26.3) |
| Longest AF duration, mo                                                                                                                                                                                                                                             | 5 (2-8)       |
| CHA <sub>2</sub> DS <sub>2</sub> -VAsC score                                                                                                                                                                                                                        | 2 (1-3)       |
| No. of previous cardioversions for AF                                                                                                                                                                                                                               | 1.7 ± 1.6     |
| Long-standing persistent AF                                                                                                                                                                                                                                         | 51 (18.1)     |
| AAD at time of enrollment                                                                                                                                                                                                                                           | 224 (79.7)    |
| Flecainide                                                                                                                                                                                                                                                          | 52 (18.5)     |
| Sotalol                                                                                                                                                                                                                                                             | 73 (26.0)     |
| Amiodarone                                                                                                                                                                                                                                                          | 99 (35.2)     |
| Values are mean ± SD, n (%), or median (Q1-Q3). <sup>a</sup> Defined as >8 standard drinks per week.                                                                                                                                                                |               |
| AAD = antiarrhythmic drug; AF = atrial fibrillation; BMI = body mass index; E/e' = ratio of early diastolic transmitral flow velocity to early diastolic mitral annular velocity; LAVI = left atrial volume index; LVEDD = left ventricular end-diastolic diameter. |               |

A two-tailed *P* value <0.05 was considered statistically significant.

## RESULTS

Patient selection and follow-up for this substudy are outlined in [Figure 1](#). Baseline LAVI data and 12-month follow-up data were available for 281 (83.1%) of 338 patients in the CAPLA study. There was no significant difference in characteristics comparing patients with and without baseline LAVI data ([Supplemental Table 1](#)). Additional arrhythmia recurrence data at 3 years were available for 276 patients (81.7%), with attrition due to 4 deaths and 1 patient lost to follow-up.

**BASELINE CHARACTERISTICS.** Baseline characteristics for this cohort are shown in [Table 1](#). Mean age was 64.8 ± 9.8 years and 77.6% were male. All patients had persistent AF at enrollment, with a median

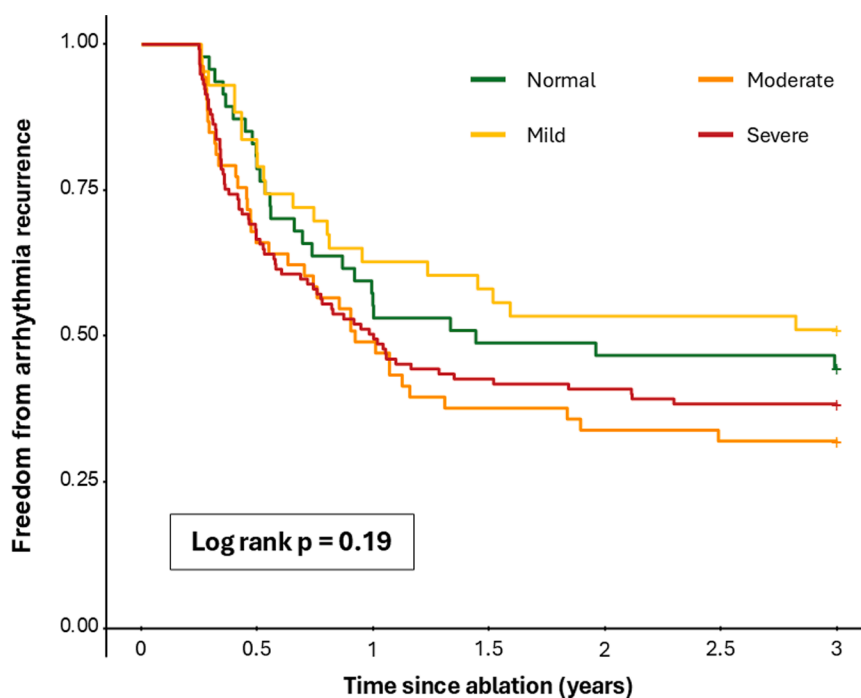
time since AF diagnosis of 13 months (Q1-Q3: 6.3-26.3 months); the median duration of continuous AF was 5 months (Q1-Q3: 2-8 months). Mean LAVI across the cohort was 47.9 ± 16.4 mL/m<sup>2</sup>. Based on standard thresholds, LA volume was normal in 50 (17.8%) patients, mildly dilated in 44 (15.7%), moderately dilated in 69 (24.6%), and severely dilated in 118 (42.0%). Mean baseline LVEF was 52.6% ± 12.3%, with 80 patients (28.5%) exhibiting LVEF <50%. Patients with LVEF <50% had significantly higher baseline LAVI compared with those with LVEF ≥50% (54.6 ± 18.2 mL/m<sup>2</sup> vs 45.6 ± 14.3 mL/m<sup>2</sup>; *P* < 0.001).

Addition of PWI was randomly assigned as part of the CAPLA study design and was used in 151 (53.7%) of 281 patients in this study. Addition of PWI was balanced across LAVI strata (*P* = 0.52) ([Supplemental Table 2](#)).

**ARRHYTHMIA RECURRENCE.** At 12 months' post-ablation, atrial arrhythmia recurrence occurred in 130 (46.3%) of 281 patients, increasing to 173 patients (61.6%) at 3 years. Kaplan-Meier curves stratified according to LAVI category (normal, mild, moderate, and severe) revealed no significant differences in arrhythmia-free survival over 3 years of follow-up (log-rank test, *P* = 0.19) ([Figure 2](#)). A collapsed analysis comparing any LAVI enlargement (mild to severe) vs normal LAVI likewise exhibited no significantly increased risk of arrhythmia recurrence (HR: 1.25; 95% CI: 0.72-2.12; *P* = 0.43) ([Figure 3](#)). In a multivariable logistic regression analysis at 12 months, LAVI was not independently associated with recurrence after adjusting for age, sex, LVEF, body mass index, AF duration, hypertension, and obstructive sleep apnea (OR: 1.01; 95% CI: 0.99-1.02; *P* = 0.55).

Redo ablation was performed in 87 (31.0%) of 281 patients, with similar rates across LA size strata (30.0% normal, 22.7% mild, 34.8% moderate, and 32.2% severe LA enlargement; *P* = 0.57) ([Supplemental Table 2](#)). Multiple redo ablation procedures were only performed in 16 patients (5.7%), and rates of multiple redo did not differ across LA size groups (*P* = 0.93).

**ARRHYTHMIA BURDEN.** Atrial arrhythmia burden data were available in all 281 patients at 1 year and in 225 (80.1%) patients at 3 years' postablation. Twice-daily plus symptomatic TTM recordings were used in 206 patients (73%), with a total of 81,572 TTM ECG recordings analyzed (median of 397 recordings per patient; Q1-Q3: 351-448). Implanted cardiac devices for continuous rhythm surveillance were used in 42 patients (15%), and ambulatory ECG monitoring was used in the remaining 33 patients (12%). Twelve-month AF burden was similar across categories of LA dilation severity, with values of 0% (Q1-Q3: 0%-1.1%)

**FIGURE 2** Time to Arrhythmia Recurrence Stratified According to LAVI Category

Kaplan-Meier curves depicting freedom from atrial arrhythmia recurrence after first-time ablation, stratified according to baseline left atrial volume index (LAVI) category (normal,  $\leq 34$  mL/m<sup>2</sup>; mild, 34.1–42 mL/m<sup>2</sup>; moderate, 42.1–48 mL/m<sup>2</sup>; and severe,  $>48$  mL/m<sup>2</sup>). No significant differences were observed among categories over 3-year follow-up (log-rank test,  $P = 0.19$ ).

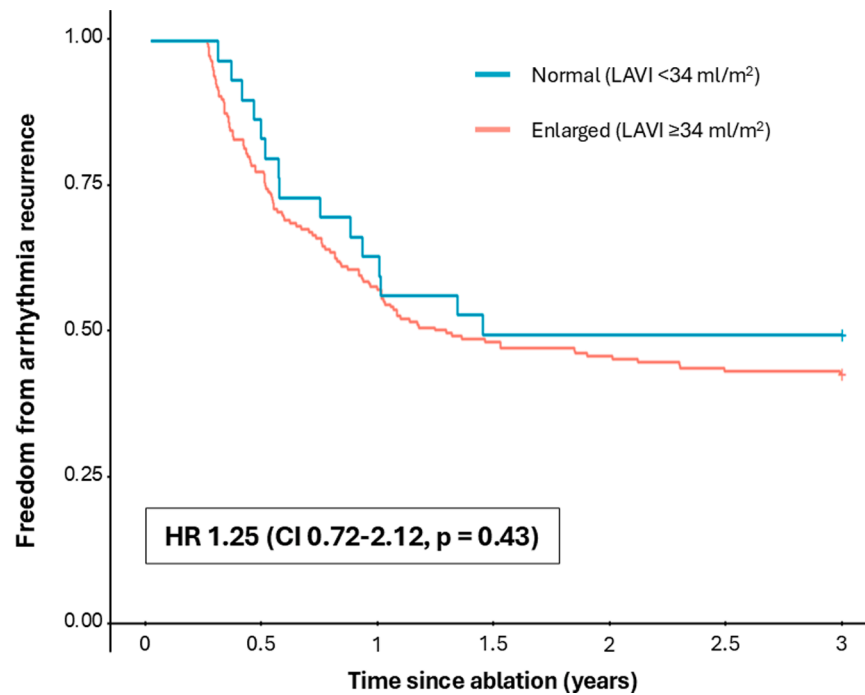
in the normal group, 0% (Q1-Q3: 0%-1.8%) in the mild group, 0% (Q1-Q3: 0%-1.1%) in the moderate group, and 0% (Q1-Q3: 0%-8.4%) in the severe group at 1 year (Table 2). When the analysis was restricted to patients with non-zero arrhythmia burden, clearer gradients emerged. At 1 year, median AF burden increased progressively with LA dilation severity, from 1.8% (Q1-Q3: 0%-5.2%) in the normal LA size group to 8.7% (Q1-Q3: 0%-30.0%) in the severe LA dilation group ( $P = 0.03$ ) (Central Illustration).

The relationship between baseline LAVI as a continuous variable and atrial arrhythmia burden was further explored by using linear regression analysis. At 1 year postablation, higher LAVI yielded a modest but statistically significant association with greater AF burden ( $R^2 = 0.12$ ;  $P < 0.001$ ) (Figure 4). However, the relationship between LAVI and AF burden was attenuated over longer term follow-up, with no significant association observed at 3 years ( $R^2 = 0.01$ ;  $P = 0.49$ ).

Finally, we evaluated the relationship between baseline LA size and phenotype of arrhythmia recurrence (paroxysmal vs persistent). Among 124

patients with AF recurrence at 1 year in whom recurrence phenotype was known, 76 (61.3%) had paroxysmal AF recurrence, whereas 48 (38.7%) had persistent AF recurrence. Higher LAVI was significantly associated with increased odds of persistent rather than paroxysmal recurrence (OR: 1.78 per 10 mL/m<sup>2</sup>; 95% CI: 1.33–2.47;  $P < 0.001$ ).

**QUALITY OF LIFE.** AFEQT scores at pre-ablation, 12 months' postablation, and 3 years' postablation stratified according to LA size are shown in Table 3. AFEQT scores were similar at baseline ( $P = 0.53$ ). Marked improvements were observed at 12 months in all groups, with mean scores increasing by 28.5 points in normal atria, 30.9 points in mild dilation, 28.6 points in moderate dilation, and 33.6 points in severe dilation. These gains were sustained at 3 years, with absolute improvements from baseline of 35.3, 36.3, 35.9, and 38.3 points, respectively. Importantly, there were no significant differences in the magnitude of improvement between groups at either 12 months' ( $P = 0.76$ ) or 3 years' ( $P = 0.44$ ) postablation.

**FIGURE 3** Time to Arrhythmia Recurrence: Normal vs Enlarged Left Atrium

Kaplan-Meier curves comparing arrhythmia-free survival between patients with a normal (left atrial volume index [LAVI] <34 mL/m<sup>2</sup>) and an enlarged (LAVI ≥34 mL/m<sup>2</sup>) left atria. No significant difference in recurrence-free survival was observed (HR: 1.25; 95% CI: 0.72-2.12; *P* = 0.43).

## DISCUSSION

The central finding of the current study is that baseline LAVI was not predictive of binary atrial arrhythmia recurrence after catheter ablation for persistent AF at 1 and 3 years' follow-up. This was consistent across both Kaplan-Meier, Cox, and logistic regression analyses. Important additional findings from our cohort include the following: 1) among patients who experienced arrhythmia recurrence, higher baseline LAVI was

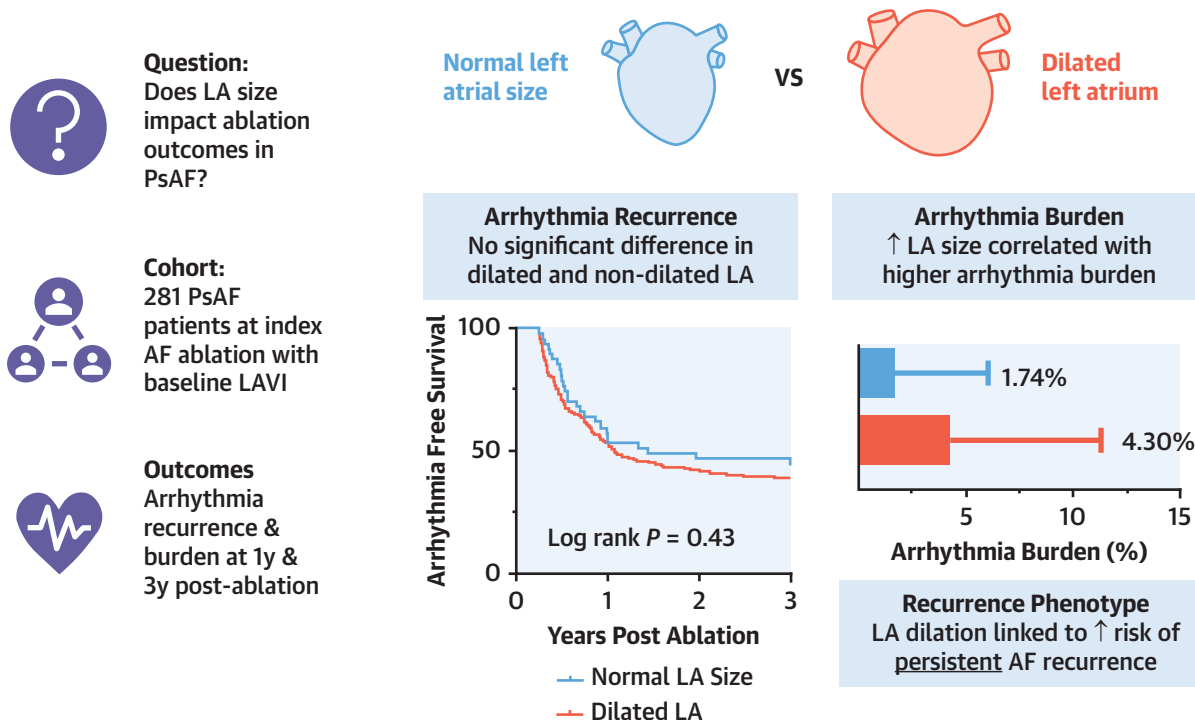
predictive of higher arrhythmia burden at 1 year; 2) higher LAVI was associated with increased odds of persistent rather than paroxysmal recurrence; and the 3) low median AF burden and excellent AF-related quality of life were achieved even among patients with severely enlarged baseline LA size.

**LA DILATION AND INCIDENT AF.** LA enlargement is a well-recognized risk factor for the development of AF. This is highlighted in the 2023 American Heart Association/American College of Cardiology/Heart

**TABLE 2** Atrial Arrhythmia Burden at 12 Months' and 3 Years' Postablation, Stratified According to LA Size

|                           | Arrhythmia Burden |                 |                 |                 | <i>P</i> Value          |
|---------------------------|-------------------|-----------------|-----------------|-----------------|-------------------------|
|                           | Normal LA         | Mild LAE        | Moderate LAE    | Severe LAE      |                         |
| 12 mo                     |                   |                 |                 |                 |                         |
| Overall (N = 281)         | 0% (0%-1.1%)      | 0% (0%-1.8%)    | 0% (0%-1.1%)    | 0% (0%-8.4%)    | 0.23                    |
| Non-zero burden (n = 127) | 1.8% (0%-5.2%)    | 6.0% (0%-13.7%) | 2.0% (0%-4.9%)  | 8.7% (0%-30.0%) | <b>0.03<sup>a</sup></b> |
| 3 y                       |                   |                 |                 |                 |                         |
| Overall (n = 225)         | 0% (0%-0%)        | 0% (0%-0%)      | 0% (0%-0.8%)    | 0% (0%-2.7%)    | 0.15                    |
| Non-zero burden (n = 69)  | 9.0% (0%-10.2%)   | 6.4% (0%-23.7%) | 7.3% (0%-18.3%) | 6.3% (0%-37.9%) | 0.51                    |

Values are median (Q1-Q3). Overall arrhythmia burden and non-zero arrhythmia burden are shown separately. Normal left atrial (LA) size (≤34 mL/m<sup>2</sup>), mild LA enlargement (LAE) (34.1-42 mL/m<sup>2</sup>), moderate LAE (42.1-48 mL/m<sup>2</sup>), or severe LAE (>48 mL/m<sup>2</sup>). <sup>a</sup>For value in **boldface**, *P* < 0.05.

**CENTRAL ILLUSTRATION** Left Atrial Size and Outcomes in Persistent Atrial Fibrillation

William JB, et al. JACC Clin Electrophysiol. 2026;■(■):■-■.

Impact of left atrial dilation on ablation outcomes in persistent atrial fibrillation. AF = atrial fibrillation; LA = left atrial; LAVI = left atrial volume index; PsAF = persistent atrial fibrillation.

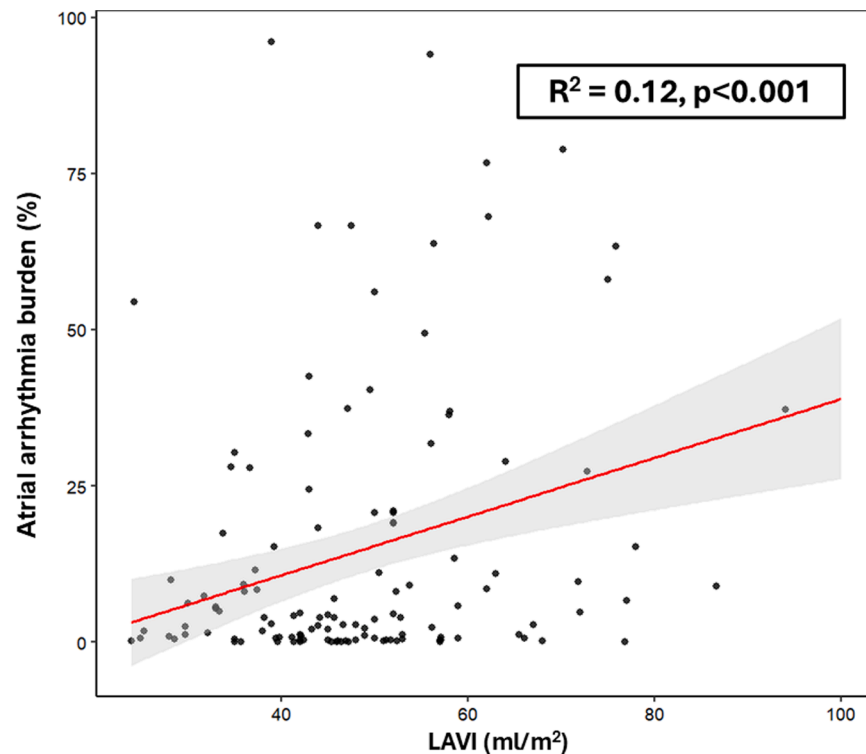
Rhythm Society guidelines, which identify both LA size and impaired atrial function as imaging markers of incident AF.<sup>11</sup> LA enlargement is frequently observed in patients referred for AF catheter ablation, with prior reports suggesting a prevalence of 60% in paroxysmal AF,<sup>12</sup> rising to >85% with persistent AF.<sup>13</sup> In the current cohort, LA dilation was present in 82.2% of patients, consistent with these previous findings.

LA enlargement and AF are closely linked in a bidirectional relationship. Dilation of the left atrium due to chronic LA pressure overload promotes the formation of interstitial fibrosis, mediated through remodeling pathways such as endothelin-1 signaling.<sup>14</sup> Atrial fibrosis in turn leads to conduction heterogeneity, localized re-entry, and wavefront dispersion, all of which contribute to the maintenance of AF.<sup>15,16</sup> Sustained episodes of AF have also been shown to trigger both structural and electrical atrial remodeling, characterized by impaired contractile and reservoir LA function,<sup>17</sup> shortening of the effective refractory period,<sup>18</sup> and dysregulated

calcium handling,<sup>19</sup> ultimately worsening LA dilation. Together, these processes establish a self-perpetuating cycle of pathologic remodeling and the landmark observation of “AF begets AF.”<sup>20</sup>

**RELATIONSHIP WITH ARRHYTHMIA RECURRENCE.** LA size has long been conceptualized as an important prognostic marker to guide ablation candidacy. In the 2024 European Heart Rhythm Association/Heart Rhythm Society consensus statement on catheter and surgical ablation for AF, it is advised that LA size should be integrated into clinical decision-making when identifying patients suitable for ablation.<sup>21</sup> Despite this recommendation, existing data have predominantly centered on patients with paroxysmal AF with short-term follow-up. In a 2018 meta-analysis comprising 3,822 patients across 21 studies, LAVI exhibited a modest association with AF recurrence, with each 1 unit increase in LAVI increasing recurrence risk by 3%.<sup>5</sup> Notably, approximately 70% of patients in this meta-analysis had paroxysmal AF. Furthermore, follow-up duration



**FIGURE 4** Baseline LAVI and 12-Month Atrial Arrhythmia Burden

Scatter plot showing the relationship between baseline left atrial volume index (LAVI) and atrial arrhythmia burden at 12 months' post-ablation. Higher baseline LAVI was modestly associated with greater arrhythmia burden ( $R^2 = 0.12$ ;  $P < 0.001$ ).

across the included studies was highly variable, often limited to 12 months but as low as 6 months. Hence, robust evidence examining the prognostic value of LA size in persistent AF ablation and long-term outcomes is lacking.

In the CAPLA cohort, which included all comers with persistent AF irrespective of LA size, baseline LAVI was not predictive of binary arrhythmia recurrence at either 1 or 3 years when examined

across categorical thresholds of normal, mild, moderate, and severe dilation. This finding aligns with other studies questioning the utility of LA size as a standalone predictor of ablation success in persistent AF. In a small, single-center study involving 79 patients with persistent AF, LA volume was not predictive of arrhythmia recurrence after catheter ablation.<sup>22</sup> In the CAMERA-MRI (**Catheter Ablation Versus Medical Rate Control in Atrial**

**TABLE 3** AFEQT Score at Each Time Point Postablation, Stratified According to LA Size

|                          | AFEQT Score |             |              |             | P Value |
|--------------------------|-------------|-------------|--------------|-------------|---------|
|                          | Normal LA   | Mild LAE    | Moderate LAE | Severe LAE  |         |
| Pre-ablation             | 57.6 ± 23   | 56 ± 24.5   | 51.2 ± 22.5  | 52.5 ± 22.5 | 0.53    |
| 12 mo                    | 80.4 ± 20.3 | 81.5 ± 20.1 | 74.8 ± 22.5  | 78.7 ± 19.4 | 0.65    |
| 3 y                      | 88.3 ± 14.6 | 92.3 ± 8.9  | 87.6 ± 16.3  | 87.2 ± 16.5 | 0.51    |
| Δ (preablation to 12 mo) | 28.5 ± 22.1 | 30.9 ± 20.9 | 28.6 ± 20.5  | 33.6 ± 19.5 | 0.76    |
| Δ (preablation to 3 y)   | 35.3 ± 20   | 36.3 ± 20.5 | 35.9 ± 20.5  | 38.3 ± 20.9 | 0.44    |

Values are mean ± SD. Normal LA size ( $\leq 34$  mL/m<sup>2</sup>), mild LAE (34.1–42 mL/m<sup>2</sup>), moderate LAE (42.1–48 mL/m<sup>2</sup>), or severe LAE ( $>48$  mL/m<sup>2</sup>).

AFEQT = Atrial Fibrillation Effect on Quality-of-Life questionnaire; other abbreviations as in Table 2.



**Fibrillation and Systolic Dysfunction)** study examining patients with persistent AF and concomitant LV systolic dysfunction, LA volume was not associated with arrhythmia recurrence.<sup>23</sup> Likewise, a recent analysis by Georgi et al<sup>24</sup> showed that low-voltage LA areas, but not LA volume, predicted recurrence after redo ablation. Taken together, these studies suggest that LA enlargement carries less prognostic significance in the setting of persistent AF ablation.

**RELATIONSHIP WITH ARRHYTHMIA BURDEN.** Arrhythmia burden is increasingly recognized as a valuable outcome measure after catheter ablation, correlating with both quality of life and health care utilization.<sup>25,26</sup> In the current study, although baseline LAVI did not predict binary recurrence, larger LA size was associated with greater arrhythmia burden and a higher likelihood of persistent recurrence at 12 months' postablation. This is a novel and important finding, for which there are several possible explanations. First, increased LA size may not be predictive of pulmonary vein reconnection, which is often the dominant mechanism of binary arrhythmia recurrence after ablation. Second, LA dilation may be associated with adverse electrical remodeling and fibrosis, which perpetuate longer duration arrhythmia relapses once recurrence occurs.

However, pre-ablation LA size did not predict atrial arrhythmia burden at 3 years' postablation. This may reflect the cumulative influence of multiple factors beyond baseline LA size, including progressive atrial remodeling, repeat ablation procedures, influence of antiarrhythmic drug therapy, less intensive monitoring, or the emergence of comorbid disease. Over longer follow-up, these evolving processes may dilute the prognostic impact of a single baseline echocardiographic measurement. This finding emphasizes that LA size is not a static risk marker but interacts dynamically with disease progression and treatment pathways.

We also found that increased LA size predicted a persistent pattern of recurrence after ablation. Recent studies have emphasized that AF phenotype can shift after catheter ablation, with many patients who relapse after ablation for persistent AF exhibiting paroxysmal rather than persistent recurrences.<sup>26,27</sup> The observation that LA dilation is linked to a higher likelihood of persistent relapse supports the concept that atrial enlargement reflects a greater burden of fibrosis, which facilitates longer AF episodes and promotes arrhythmia maintenance.

Despite the association between LA dilation and greater arrhythmia burden, it is important to note

that median overall arrhythmia burden at both 12 months' and 3 years' postablation was very low (0%) across all strata of LA dilation. Moreover, sustained improvements in AF-related quality of life were observed in all groups, including those with severe LA dilation pre-ablation. Taken together, these findings suggest that LA size has limited utility as a predictor of ablation success in persistent AF and that reliance on LAVI alone may unnecessarily exclude patients who stand to derive substantial symptomatic benefit from catheter ablation.

**STUDY LIMITATIONS.** First, although derived from a large randomized trial, the current analysis was post hoc and therefore subject to residual confounding. Second, LA size was assessed by echocardiography at a single time point, without incorporation of advanced imaging modalities such as **cardiac magnetic resonance or computed tomography imaging**, which may more accurately characterize atrial remodeling. Third, arrhythmia recurrence and burden was adjudicated by using heterogeneous monitoring techniques, including both continuous (implantable loop recorders) and intermittent (twice-daily transtelephonic ECG) methods.

## CONCLUSIONS

Baseline LA size was not predictive of arrhythmia recurrence at 1 or 3 years after ablation for persistent AF. Although large LA size was associated with greater arrhythmia burden and more persistent relapse at 1 year, this relationship was attenuated over longer term follow-up. Low AF burden was observed across all strata of LA size, even in patients with severe enlargement, indicating that LA size may have limited value in selecting candidates for AF ablation.

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## PERSPECTIVES

**COMPETENCY IN MEDICAL KNOWLEDGE:** LA enlargement is a common finding in patients with persistent AF.

**COMPETENCY IN PATIENT CARE:** Patients with persistent AF and marked LA enlargement can still achieve excellent rhythm control and sustained quality-of-life improvements after catheter ablation.

**TRANSLATIONAL OUTLOOK 1:** The relationship between LA size and AF burden highlights that structural remodeling influences the phenotype and burden of arrhythmia recurrence rather than its presence alone.

**TRANSLATIONAL OUTLOOK 2:** Future studies should integrate structural, electrical, and clinical predictors to refine ablation selection criteria and tailor postablation monitoring strategies for patients with persistent AF.

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**KEY WORDS** catheter ablation, left atrial size, persistent atrial fibrillation

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**APPENDIX** For supplemental tables, please see the online version of this article.